

Table 9.2 Comparison of Optimization Techniques

Optimization Technique	Advantages	Disadvantages
Non-Linear Optimization	Determines exact solution to the exact problem	Takes too long to solve to be useful to traders
	Takes full advantage of diversification and hedging	Many parameters—one for each stock and period
Quadratic Optimization	Provides most accurate trade schedule	Many parameters—one for each stock and each period
	Takes full advantage of diversification and hedging	Slow solution for larger trade lists
		Could require multiple iterations
Trade Schedule Exponential	Very fast optimization solution	Does not allow full freedom in specifying trade schedule
	Few parameters—one per stock	Trade schedule is forced to follow exponential decay
	Takes very good advantage of diversification and hedging	
	Very accurate model	
Residual Schedule Exponential	Very fast optimization solution	Does not allow full freedom in specifying trade schedule
	Few parameters—one per stock	Forces a front-loaded trade schedule
	Takes very good advantage of diversification and hedging	
	Very accurate model	
Trade Rate	Quickest optimization solution	Does not take full advantage of diversification and hedging
	Adapts to changing market conditions in real-time	Requires approximation of residual risk function
	Few parameters—one per stock	Least accurate of the methods

Risk Aversion. The risk aversion parameters were randomly selected from the following values $\lambda = 0.3, 1, 2$.

Trading Days. We broke the day into thirteen intervals of equal volume.

In total, our simulation experiment took several days to run. The optimizations were run using a 64-bit pc, with an Intel i7 processor, 2.6 GHz, and with 8 Gigs of RAM. Since the actual optimization times are also